

EXPERIENCE GAINED FROM THE OPERATION OF THE BR-5 REACTOR, 1964-5

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1. Introduction

Details of the main results acquired after 5 years' operation of the BR-5 reactor (1959-63) were given in a report presented at the Third International Conference of the United Nations on the Peaceful Uses of Atomic Energy.

The present report describes the results obtained from the operation of the BR-5 reactor with an active zone of plutonium oxide, start-up and the studies conducted on the monocarbide variant of the reactor charge. After March 1962 the charge in the BR-5 active zone was replaced by plutonium dioxide and uranium dioxide assemblies. For further operation the reactor was loaded with 80% plutonium dioxide assemblies from the original charge with a maximum burnup of 4.85% and uranium dioxide assemblies of similar design.

PuO₂ assemblies were concentrated in the central part of the active zone and during loading they were rearranged in order to equalize their burnup during subsequent operation of the reactor.

In November 1964, owing to a major increase in the release of gaseous and solid fission products to the circuit as a result of further leaks in the plutonium dioxide fuel elements, the BR-5 was shut down to enable monocarbide fuel to be loaded in the active zone. At the time when the reactor was shut down for reloading, the maximum burnup in the plutonium dioxide fuel elements forming part of the original charge was 6.5% and that of the 90%-enriched uranium dioxide elements 1.4%. At this time the fuel elements in some of the uranium monocarbide assemblies, which were placed in the reactor in 1964 in order to study their efficiency, had reached a total burnup of 0.8%.

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The loading of the active zone of the reactor was followed by the completion of maintenance work, which was mainly directed at the replacement of leaking drainage valves in the primary circuit loops, the replacement of cable in various control systems associated with the main circuits, owing to its unsatisfactory electrical insulating properties, and the attachment of additional experimental devices to the reactor. After this, in May 1965, the reactor went into operation with a monocarbide fuel in the form of UC; the characteristics of the active zone with the new fuel charge were studied, and in June 1965 the BR-5 was stepped up to its full design power of 5000 kW.

The primary coolant was not changed when the reactor charge was altered, and the primary circuit was purged of radioactive contamination by means of sodium oxide cold traps before the maintenance work was carried out.

2. In-pile Studies of the Contamination of the Primary Circuit by Fission Products

During the operation of sodium cooled fast industrial reactors there is a possibility that leaks might develop in the cladding of the various fuel elements and that the fuel therefore might come in contact with the coolant. A question which assumes great practical importance here is that of the contamination of the reactor circuit by fission products when the installation is operating with defective fuel elements. The main purpose of the operation of the BR-5 reactor during the period 1962-4 was as follows: to study the contamination of the coolant, the walls of the flow equipment, and the gas in the reactor's gas blankets by solid and gaseous fission products; to examine the retention, by means of sodium oxide cold traps, of radioactive fragments entering the circuit when the active zone contains fuel assemblies which are no longer leak-tight; and also to obtain further information on the high burnup of fuel in plutonium dioxide elements.

Power operation of the reactor during the specified period was subordinated to the completion of the programme of studies on contamination and the cleaning of the primary circuit. The aim here was that, as the studies progressed, no major difficulties should be caused in the operation of the primary circuit as a result of the contamination of the coolant, the walls of the equipment and the pipework because of radioactive fission products.

During 1962-3 the BR-5 operated at 20 and 40% of its rated power. In 1964 the output was stepped up to 3000 kW, and then to 3500 kW (70% of rated power). As a result of 3 years' operation of the reactor with fuel elements which had reached high burnup and had leaks in the cladding, it was possible to acquire some experience and information about the contamination of the coolant, the inert gas and the primary circuit equipment because of radioactive fission products. It was found that the release of

gaseous and volatile solid fission products from the fuel elements to the coolant was mainly dependent on the reactor power, and hence on the temperature of the fuel elements. At a power of 1000 kW, for example, there was virtually no release of fission products from the fuel elements, and the contamination of the circuit was found to consist of the isotope caesium 137, the activity of which in the coolant was $\sim 0.3 \mu\text{c/g}$ of sodium, and which was due to the residual contamination of the primary coolant and also to the fact that the quantity of caesium removed from the sodium by the cold trap was extremely small. At a reactor power level of 2000-3500 kW, the release of fragments increased considerably, and the radioactive contamination of the primary circuit, after the decay of the sodium 24, consisted of relatively short-lived fission products, iodine 131, barium 140, lanthanum 140, caesium 136 and caesium 137. The activity of the iodine 131 was greater than that of the other isotopes by several factors, amounting to $30 \mu\text{c/g}$ of sodium. At the same time it was found that the amount of iodine 131 released from the leaking fuel elements was usually lower than that of the caesium 137 released. Fission fragments such as iodine 131, caesium 136 and caesium 137 are efficiently retained by the oxide cold trap. For example, a sodium oxide cold trap which had been in use for 2 years was cut out of the circuit in the middle of 1964 and the total activity of the caesium 137 was found to be ten times that of the caesium present in the remainder of the primary circuit at that time.

The trap does not possess any great efficiency with regard to the retention of such isotopes as zirconium 95, niobium 95, barium 140 and lanthanum 140, and these radioactive products are mainly deposited on the walls of the equipment and the piping. In 1964, when the reactor was running at 3500 kW, special studies were performed to examine the influence of the cold trap on the activity of the fission products in the coolant and on the walls of the primary circuit piping. Measurements carried out on the by-pass section of the primary circuit with the aid of a gamma-ray scintillation spectrometer showed that when the cold trap was working the activity of the coolant and the pipe walls attributable to caesium 137 and iodine 131 was only one-tenth of that when it was not in use.

The activity of the gaseous fission products in the gas spaces is markedly dependent on the power level. At a power of 1000 kW the xenon 133 and xenon 135 activity was of the order of a few tenths of a mc/l, but at 3000 kW it amounted to 20 and 5 mc, and at 3500 kW to 200 and 50 mc/l respectively. Despite the different conditions for the release of gas from the sodium to the gas spaces in the circuit (pumps, central reactor pipe), the average xenon 133 activity in the gas blankets is roughly the same, while the ratio of xenon 133 to xenon 135 in them is quite different.

At a fairly high iodine 131 content in the coolant and on the walls of the flow equipment, the release of iodine 131 to the gas spaces in the primary

circuit is not very marked. The iodine 131 activity in the gas did not exceed 10^{-2} mc/l, which is lower than the xenon 133 and xenon 135 activity by a factor of 10^3 – 10^4 .

In order to improve the available means of monitoring the activity of fission products in the coolant and on the walls of the primary circuit piping, a spectrometric by-pass section similar to the one on the first loop of the primary circuit was fitted to the second loop. By means of these two independent by-pass sections it is possible to ensure a constant coolant flow through one of them with only short intervals for measuring the activity on the walls of the section. The second section is used for measuring the activity of the coolant.

The improvements carried out to the gas sampling system enabled the activity of the gas to be monitored from the gas spaces in the central reactor pipe and both of the primary circuit pumps. In addition, one of the primary pumps was equipped with a recirculation line with a sealed gas blower, by means of which the gas from the gas space in this pump could be recirculated over the spectrometer recorder.

The measurement of the activity of the circulating gas revealed the presence of neon 23 in the inert gas in the primary circuit, formed from sodium 23 by the (*n,p*) reaction. The total activity of the neon in the gas spaces in the circuit was about 50 c at rated power.

The activity of the sodium circulating in the by-pass section was measured a few minutes after reactor power was increased (i.e. at low sodium 24 activity), as a result of which it was possible to determine the neon activity in the coolant, which was equal to 0.6 c/l (i.e. about 1000 c for the entire circuit).

3. Unloading of the Active Zone in 1965 and Leak-tightness Studies on Fuel Elements

In January 1965 the active zone was entirely unloaded during a period of 9 days and placed in the spent fuel store. Unloading was carried out by the normal technique, without cooling of the assemblies. The sodium and its oxides were not washed off the bundles removed from the active zone.

In an attempt to examine the possibility of unloading high-activity fuel assemblies into molten lead, ten assemblies which had just been removed from the active zone were immersed in casks containing molten lead at $\sim 500^\circ\text{C}$. No external signs of a reaction were observed when the sodium on the surface of the assembly dissolved in the lead. The residual sodium is washed off the assemblies when they are placed in the lead and, in addition, the source of any radioactive contamination is localized in the event of any of the fuel elements in the assembly being defective. The assemblies removed and placed in molten lead consisted of eight plutonium dioxide assemblies

with a maximum burnup of 4.2-6.2% and two uranium dioxide assemblies with a maximum burnup of 1.4 and 0.96%. The unloading of the assemblies into the molten lead presented no difficulties whatever.

A month after unloading from the active zone, a check was conducted on the leak-tightness of the cladding of the fuel elements in all the assemblies in the store. Gas was drawn off into an ionization chamber (useful volume 70 litres) through a pipe mounted in the space between the hood of the store and the bundle in it. The temperature of the assemblies was determined on the basis of the after-heat (~ 40 W per assembly) and the temperature of the surrounding air. The activity of the gas drawn off from the assemblies provided an indication of the leak-tightness of the fuel cladding. In this way, checks were carried out on fifty-nine assemblies containing plutonium dioxide elements with a burnup of up to 6.5% and twenty-one assemblies containing uranium dioxide and uranium monocarbide with a burnup of up to 1.5%. In the course of the check performed, all the uranium assemblies and thirty-two of those containing plutonium were found to be leak-tight; the activity of the gas released from them did not exceed that of the gas from the dummy assemblies containing metallurgical specimens which had been removed from the active zone and which, like the hot assemblies, had not been washed free of sodium. The activity of the gas from the seventeen plutonium assemblies with a burnup of more than 5.1% was greater than the background by a factor of 10-1000. Measurement of the gamma spectrum of the gas from the assemblies showed that the activity is made up of krypton-85 (measurements performed 4 months after shutdown of the reactor).

In order to study the laws governing the release of active gas from leaking fuel elements, tests were performed in a special hood, fitted with an electric heating element and a thermocouple, on two of the leaking assemblies and fifteen assemblies emitting background activity. In both leaking assemblies a marked correlation was observed between the gas release and the temperature of the assembly. When the assembly was heated from 40 to 600°C, the gas release rose 10^3 - 10^4 times (see Fig. 1). During the course of the tests in the storage hood, the two assemblies having activities below background level showed a very marked gas release when heated up to 500°C.

Heating to 700°C revealed a high activity in the majority of assemblies. The results of the study of gas release from the fuel elements in the assemblies which were tested in the special heated hood are given in Table 1.

In order to be certain that no irreversible changes occurred in the fuel claddings as a result of heating to 700°C, two of the seven assemblies which had shown a marked activity after heating to 700°C were retested at 40 and 400°C after cooling, no major gas release being observed.

The studies on the activity of the gas drawn off from the casks containing solidified lead and the plutonium assemblies with defective cladding showed that gaseous fission products cannot pass through solidified lead. When the

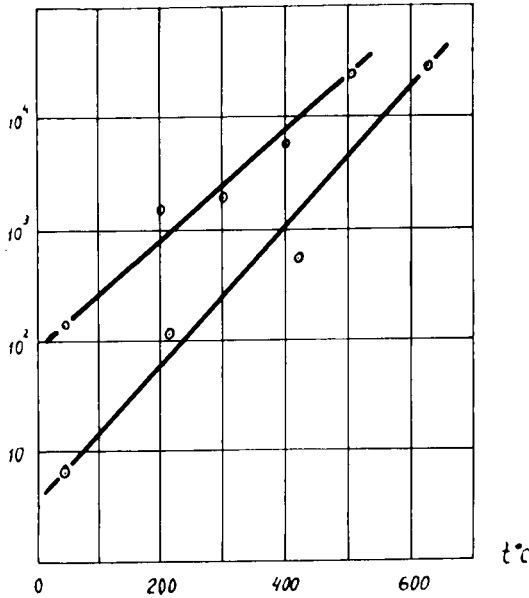


FIG. 1. Active gas release from defective fuel elements as a function of the temperature.

TABLE 1

No.	No. of assembly	Relative gas activity from assemblies			
		Test in store hood	Tests in special airtight hood at different temperatures		
			50°C	500°C	700°C
1	54	0.2	3.5	20	Not tested
2	70	0.7	4	8	Not tested
3	74	0.1	6	17	Not tested
4	24	0.5	13	8,000	Not tested
5	76	0.2	40	>20,000	Not tested
6	11	0.3	1.3	6	4
7	16	0.3	2	6	16
8	21	0.2	10	10	15
9	9	0.2	2	8	>20,000
10	15	0.4	4	8	>20,000
11	22	0.3	1	30	>20,000
12	35	0.2	5	5	>20,000
13	41	0.7	1	14	>20,000
14	42	0.2	3	6	>20,000
15	75	0.3	1	14	>20,000

lead is melted, gaseous fission products can pass through it freely. The activity of the gas released from the leaking assembly is roughly the same before loading into the lead, in the molten lead and after removal.

When an assembly containing fuel elements with defective cladding is placed in the molten lead, there is considerable contamination of the lead in the form of caesium 137. The activity of 5 kg of lead left in the cask after an assembly containing defective fuel elements had been removed was ~ 1.5 c (caesium 137 activity in the assembly ~ 120 c). The activity of the lead due to the dissolving of contaminated sodium when an assembly containing leak-tight fuel elements is placed in it is lower by a factor of 120.

Metallurgical studies were performed on four hot assemblies from the BR-5 reactor having various plutonium burnups. Examinations were carried out of the state, dimensions and leak-tightness of the fuel elements, and the amount of gas in the fuel elements and in the plutonium dioxide blocks was determined. In addition, metallographic and radiographic studies were conducted on the fuel element materials. Some of the results of the studies and the initial data for the assemblies are given in Table 2. After visual inspection, the assemblies were cut up on a remote-control milling machine and all the fuel elements carefully washed in spirit.

An external examination revealed that there were no visible signs of fracture in any of the fuel elements taken from the assembly with burnups of 2.35, 3.5 and 6.3%. There were two failed elements in the assembly with a burnup of 5.76%. The surface of the intact elements was clean, without any visible signs of deformation, and they were of a matt metallic hue throughout their length. No appreciable traces of any interaction between the cladding material and the sodium coolant were observed. The spacer wire on the surface of the elements was in satisfactory condition. Both of the defective fuel elements from the 5.76% burnup assembly had failed, and open longitudinal cracks of varying width were visible along almost the entire length of the elements (Fig. 2). The fractures in the cladding were brittle along the austenite grain boundaries without any noticeable traces of plastic deformation of the material (Fig. 3).

There is an increase in the diameter of the fuel elements, the greatest increases being observed in the middle and somewhat above the centre. The magnitude of the diameter change is a function of the plutonium burnup. For example, the average maximum increase in the diameter of elements from the assembly with a burnup of 2.35% is 0.1 mm, but 0.15 mm in the case of the elements with a burnup of 5.76%. Fuel elements with burnups of 5.76 and 6.3% show the greatest diameter (5.17-5.20 mm). The data for the changes in the fuel element diameters were used to calculate the swelling of the plutonium dioxide blocks, allowances being made for the clearance between the cladding and the blocks.

The results of this calculation are shown in Table 2, from which it can be

TABLE 2

No.	Average burnup (as %) (calc)	Integral neutron flux (n/cm ²)	Quantity of gas in ele- ment (cm ³)	Total quantity of gas (cm ³)	Relative gas release (as %)	Average max dia. in- crease (mm)	Average max dia. increase of blocks (%)	
							Total	1% burnup
1	2.35	0.98 10 ²²	8.4	19.0	44.0	0.1	2.8-3.5	1.2-1.5
2	3.50	1.5 10 ²²	17.2	31.0	55.0	0.08	3.4	1.0
3	5.76	2.4 10 ²²	27.5	46.0	60.0	0.15	3.6-5.0	0.6-0.9
4	6.3	2.7 10 ²²	—	—	—	0.12	3.5-8.0	0.6-1.3

seen that the average maximum swelling of the blocks over their diameter is about 1% for a plutonium burnup of 1%. The average swelling of the blocks over their cross-sectional area is 1.5% for a burnup of 1%.

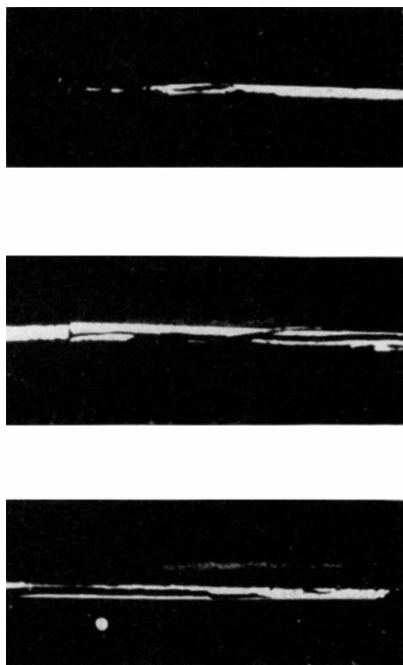


FIG. 2. Photograph of a defective fuel element from the assembly with a burnup of 5.76%: (a) top of element, (b) middle of element, (c) bottom of element.

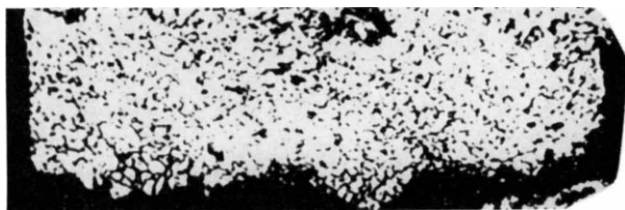


FIG. 3. Microstructure of fuel element cladding material near fracture.

A total of five to ten elements from each assembly were tested for leak-tightness. All the elements tested were found to be leak-tight. The gas was collected from the fuel elements by puncturing the cladding *in vacuo* and under a layer of liquid, four to six elements being tested from each assembly. The

results of the experiment are shown in Table 2. In addition, Table 2 contains the data on the quantity of gas in the plutonium dioxide blocks, determined

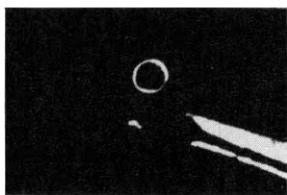
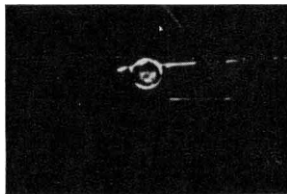


FIG. 4. Micrograph of cross-section of middle of fuel element: (a) 3.35% burnup; (b) 5.76% burnup.

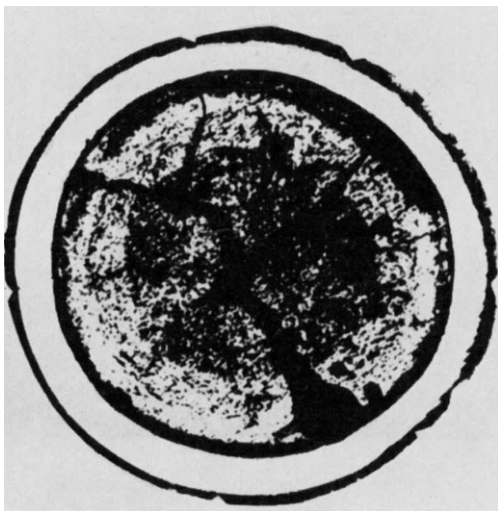


FIG. 5. Micrograph of a cross-section of a fuel element with a burnup of 5.76%.

by chemical dissolution and radiometric analysis. It can be seen from these data that the quantity of gas in the fuel elements constantly increases with

rising burnup. The gas release from the plutonium dioxide blocks is 40-45% for burnups of less than 3% and about 60% for burnups of more than 3.5%.

Macro- and microscopic studies on cross-sections of fuel elements showed that plutonium dioxide blocks with a burnup of 3.5% and above contain radial cracks and central holes, the diameters of which increase with the plutonium burnup (Fig. 4). The plutonium dioxide is marked by pores of various sizes and certain microstructural alterations are to be seen. The density of the pores changes over the cross-section of the block (Fig. 5).

No layer of interaction is to be observed in the cladding material structure either on the sodium coolant side or on the plutonium dioxide side. The cladding material (18Cr9NiTi steel) shows increased hardness and low plasticity. In the case of fuel elements with a burnup of 5.76% the microhardness of the cladding material was 320-340 kg/mm², while the relative elongation recorded during tensile tests on circular specimens was about 0.5% (at 500°C).

No changes in the properties of the blocks were observed during the radiographic studies. A slight drop was noted in the plutonium dioxide lattice constant, amounting to 0.1% in the case of 5.76% burnup.

4. Start-up and Studies on the Characteristics of the Monocarbide Fuelled Reactor

The fuel used for the second loading of the active zone of the BR-5 reactor was uranium monocarbide with a 90% U-235 enrichment with an over-stoichiometric carbon content. The density of the uranium monocarbide was 12 g/cm³. The original dimensions of the fuel element assemblies were

TABLE 3. DESIGN DIMENSIONS OF URANIUM MONOCARBIDE FUEL ELEMENTS

Shape of assembly	hexahedral
Diameter of assembly across flats	26.1 mm
Thickness of hexahedral assembly casing	0.3 mm
Height of assembly	833 mm
Number of elements in assembly	7
External diameter of cladding	8.55 mm
Thickness of cladding	0.4 mm
Diameter of fuel core	7.65 mm
Height of fuel core	380 mm

retained, but the design was changed. Each assembly in the new active zone contained seven fuel elements with fuel cores of large diameter and height (see Table 3). The fuel in the elements is encased in stainless steel cladding. The clearance space between the fuel and the cladding is filled with helium, and stainless steel wire 0.4 mm thick is used as a spacer. A space 50 mm high was left above the fuel in the elements for the collection of gaseous fission

products. The calculated maximum temperature of the fuel at the centre of the elements is 800°C.

Start-up of the monocarbide fuelled BR-5 reactor took place in May 1965. The fuel element assemblies were loaded in the active zone of the reactor, which had previously been filled with sodium. Criticality was reached when eighty-three assemblies of uranium monocarbide fuel elements (99 kg of U-235) had been loaded, together with twenty blanket assemblies of fuel elements containing natural uranium metal of the original design. The critical mass obtained agreed fairly closely with the data acquired with the two-dimensional calculation in the eighteen-group diffusion approximation, which gave the critical mass as eighty-nine assemblies.

TABLE 4. PHYSICAL CHARACTERISTICS OF THE BR-5 REACTOR WITH MONOCARBIDE FUEL

1. Rated power	5000 kW
2. Heat generation in active zone	
Average	210 kW/l
Maximum	285 kW/l
3. Form factors for heat generation	$K_r=1.14$ $K_z=1.19$ $K_v=1.36$
4. Maximum neutron flux	4.8×10^{14} n/cm ² /s
5. Effectiveness of control elements:	
Compensating cylinder	4.1%
Blanket compensator	4.8%
Automatic controller	0.1%
6. Effectiveness of uranium monocarbide:	
Fuel element assemblies at centre	0.84%
at edge of active zone	0.36%
Natural uranium fuel assemblies at edge of active zone	0.11%
7. Effect of drainage of sodium from the active zone	-4%
8. Temperature coefficient for active zone	$-3 \times 10^{-5}/^{\circ}\text{C}$
9. Asymptotic power coefficient of reactor	$-8 \times 10^{-4}/\text{MW}$

In order to conduct further studies on fuel burnup in the reactor, four assemblies of plutonium dioxide elements were loaded which had previously been in the facility and had reached a burnup of ~6%. The leak-tightness of the assemblies was tested by heating them to 500°C. At the same time two assemblies of plutonium dioxide elements and two assemblies of uranium dioxide elements with burnups of up to 1% were inserted in the reactor. The plutonium dioxide was also used as a start-up neutron source by using the (α, n) oxygen reaction and the spontaneous fission of Pu-240.

After the fuel had been loaded, studies were carried out on the reactor characteristics. The data are given in Table 4. The form factors for heat generation over the radius and height of the active zone were measured on the basis of the gamma-activity distribution in the fuel element assemblies, which were mounted at different radii. The calculated average neutron

TABLE 5

Energy range	
10-6.5 MeV	0.54
6.5-4 MeV	3.0
4-2.5 MeV	6.9
2.5-1.4 MeV	13.0
1.4-0.8 MeV	15.3
0.8-0.4 MeV	20.8
0.4-0.2 MeV	17.8
0.2-0.1 MeV	11.9
100-46.5 KeV	7.1
46.5-21.5 KeV	2.5
21.5-10 KeV	0.8
10-2.5 KeV	0.32
2.5-0.46 KeV	0.12
0.46-0.1 KeV	0.03
0.1-0 KeV	0.011

Neutron fraction (in %)

spectrum over the active zone is given in Table 5. The effect of the drainage of sodium from the active zone was measured by emptying all the sodium from the central reactor pipe under subcritical conditions. The reactor temperature

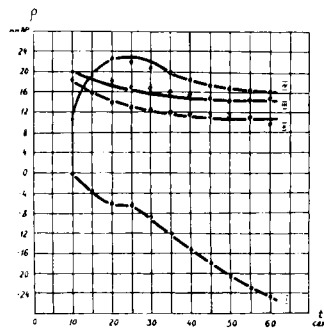


FIG. 6. Reactivity curve after power step for different sodium flow rates.
1: 240 m³/h; 2: 72 m³/h; 3: 39 m³/h; 4: 22 m³/h.
AP = automatic controller.

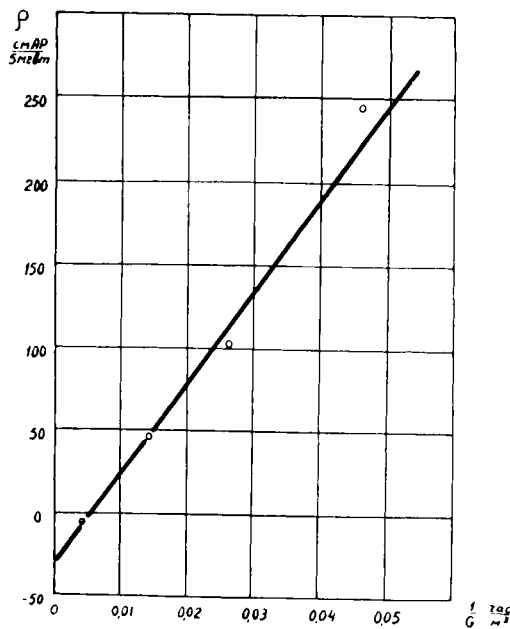


FIG. 7. Power effect as a function of the sodium flow rate.
AP = automatic controller.

coefficient, which is related to the uniform heating of the elements in the active zone and the increase in the temperature of the sodium at the reactor inlet, was measured at different power levels ranging from 10 to 4000 kW.

The temperature coefficient was found to be constant for sodium inlet temperatures ranging from 180° to 430°C. The steady state power coefficient, which is a function of the non-homogeneous heating of the reactor elements, was measured on the basis of the change in the reactivity as the power increased at constant sodium inlet temperature. The steady state power coefficient was constant throughout the power range measured. Figure 6 gives the data for the reactor power effect measured by observing the reactivity curve after the power is stepped up by 10% of the starting value for various sodium flow rates. In each measurement the inlet temperature of the sodium rose by 75°C. Figure 7 shows the power effect after 20-25 sec as a function of the reciprocal of the sodium flow. The value of the power effect for the fuel is obtained by extrapolation where $G = \infty$.

$$\alpha_{\text{fuel}} = -1.6 \times 10^{-4} / \text{MW}.$$

The reason for the negative value of the effect lies in the free elongation of the column of fuel in the new assemblies of fuel elements. At a rated sodium flow of 240 m³/h, the power effect is negative and 20-25 sec after the power surge is equal to $\alpha_n = -0.3 \times 10^{-4} / \text{MW}$. This value is made up of the negative fuel effect and the effect of the heating of the metallic structure, where the main role is played by the positive effect caused by the elongation of the central reactor pipe and the resultant displacement of the active zone relative to the regulating part of the reflector (compensating cylinder). This positive effect is clearly seen at low sodium flow rates, where the heat of the fuel is negligible.

On completion of the studies on the physical characteristics, the BR-5 reactor was raised to its rated power of 5000 kW on June 25, 1965.

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